For natural rubber (NR) and EPDM rubber composites, the vulcanization system must overcome the inherent cure incompatibility between the highly unsaturated NR and the saturated EPDM. The choice of curing agents, activators, and processing conditions is therefore critical for achieving a satisfactory co-cure and balanced final properties.

□ Vulcanization System Components and Proportions

A typical NR/EPDM blend uses a **70:30 NR:EPDM ratio** to balance NR's mechanical strength with EPDM's aging resistance . The following table outlines the key components and their typical proportions:

| Component | Function | Typical Material | Typical Loading (phr*) |

| :--- | :--- | :--- | :--- |

| **Vulcanizing Agent** | Forms crosslinks | Sulfur; Organic Peroxide (e.g., Dicumyl Peroxide) | 1.5–4.0 total |

| **Accelerator** | Speeds up cure | **TBBS** (sulfenamide), **MBTS** (thiazole), CBS, TMTD | 0.5–1.5 |

| **Activator** | Increases accelerator efficiency | Zinc Oxide (ZnO) + Stearic Acid | ZnO: 3–5; Stearic Acid: 2 |

| **Retarder** | Prevents premature scorch | PVI (N-cyclohexylthiophthalimide) | 0.15 |

*phr = parts per hundred rubber

Key Considerations for the Curing System:

* **Sulfur/Peroxide Hybrid System:** Using a combination of sulfur and a small amount of organic peroxide (e.g., 0.5–1.0 phr) has been shown to significantly improve the tensile strength and elongation at break of NR/EPDM blends compared to sulfur alone . This hybrid approach helps achieve a more successful co-cure.

* **Accelerator Selection:** **TBBS** is often the preferred choice because it provides the best processing safety and a good balance of cure between the NR and EPDM phases .

MBTS also gives good cure compatibility and high tensile strength . Using a binary accelerator system like **MBTS/TMTD** can provide the best aging resistance and blend compatibility .

□ Vulcanization Temperature and Time

The optimal cure temperature and time are determined by rheometer testing (t90 = time to 90% cure) for your specific formulation. Typical conditions for NR/EPDM blends are:

* **Temperature:** A common vulcanization temperature is **150°C** .

* **Time:** At 150°C, a typical cure time is **25 minutes** . For a 50/50 NR/EPDM blend, the optimum cure time (t90) has been reported to be around **16 minutes** .

□ Mixing Sequence

The mixing procedure is critical to prevent the NR phase from consuming the curing agents before the EPDM phase can react. A recommended approach is a multi-stage mix :

1. **First Pass:** Mix the EPDM with a portion of the activators/accelerators at elevated temperature to pre-treat it.

2. **Second Pass:** Prepare a masterbatch of NR with fillers and other additives.

3. **Third Pass:** Blend the pre-treated EPDM with the NR masterbatch.

4. **Final Pass:** Add the final curatives (sulfur, accelerators, activators) at a low temperature (<105°C) to prevent premature scorch .

If you have a specific property target (e.g., hardness, compression set) for your NR/EPDM blend, I can help you refine these starting proportions.